

Chapter 13: Visualising Solid Shapes

Worksheet 13: Visualising Solid Shapes

Learning Objective: Differentiate 2D and 3D shapes, understand nets of solids, and identify the top, front and side views of solid objects.

Worked Example:

How many faces, edges and vertices does a cube have?

Solution: A cube has 6 faces, 12 edges, and 8 vertices.

Questions (1–20)

1. Name a solid shape with no edges or vertices.
2. How many faces does a triangular prism have?
3. What is a “net” of a solid shape?
4. How many vertices does a square pyramid have?
5. Name the 3D shape with 1 curved surface and 2 flat circular faces.
6. A cube is unfolded into a net. How many squares will the net contain?
7. How many flat faces and curved surfaces does a cone have?

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8. If you look at a cube directly from the top, what 2D shape do you see?

 9. A cylinder is viewed from the side. What 2D shape does it appear as?

 10. How many edges does a triangular pyramid (tetrahedron) have?

 11. Describe the net of a cuboid with dimensions $4\text{ cm} \times 3\text{ cm} \times 2\text{ cm}$, listing the sizes of all 6 faces.

 12. A solid has a square base and four triangular faces meeting at a point. Name this solid and find its faces, edges and vertices.

 13. If a cube of side 5 cm is cut into 1 cm cubes, how many small cubes will be obtained?

 14. Identify the solid shape formed by rolling a rectangular sheet of paper into a tube.

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15. A triangular prism has two triangular faces and three rectangular faces. Find its total faces, edges and vertices.

 16. A cube-shaped box, painted on the outside, is cut into 27 equal smaller cubes (3 cuts in each direction). How many small cubes have paint on exactly 2 faces?

 17. A water tank is shaped like an upright cylinder. Describe what its top view, front view, and side view would look like.

 18. A solid is built from 5 unit cubes stacked in an L-shape (3 in a row, 2 stacked on one end). Describe its front, top and side views.

 19. Find three objects at home shaped like a cube, cylinder, and cone (e.g., a dice, a can, an ice-cream cone). Name them.

 20. Take an empty cardboard box (cuboid) and open it flat along its edges. Describe the net you get – how many rectangles do you see, and are any identical in size?

Reflection Question: Why do you think understanding nets is useful for packaging design (like cereal boxes or juice cartons)? Write your idea.

Real-Life Application Question: Look at a packaged product at home (a box, can, or tube). Identify its 3D shape and describe its faces.